Analysis of Heart Rate Variability in Elite Athletes Using ECG Data

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***Abstract*—Advancements in electrocardiogram (ECG) data analysis have enabled a deeper understanding of heart rate variability (HRV) in human hearts, distinguishing between benign cardiac adaptations and potential pathologies. This research employs wavelet transform methodologies to analyze ECG signals from Norwegian elite endurance athletes, aiming to distinguish between benign cardiac adaptations and potential pathologies. Through the detailed examination of ECG waveform features such as P-waves, QRS complexes, and T-waves, and their correlation with physiological changes due to athletic training, this study offers insights into heart rate variability (HRV). The advanced signal processing techniques utilized here not only enhance the precision of cardiovascular monitoring but also contribute significantly to the understanding and assessment of athlete health, demonstrating the critical role of sophisticated ECG data analysis in sports science and medicine.**

***Index Terms*—Electrocardiogram, Heart Rate Variability, R-peaks, RR-intervals, P-wave, QRS complex, T-wave, 12-lead configuration,**

I. Introduction & Background

Cardiovascular monitoring is a critical component in the assessment of athletic health and performance. The electrocardiogram (ECG) serves as a non-invasive and informative tool that reflects the electrical activity of the heart, providing insights into both physiological adaptations and potential pathologies. In elite athletes, extensive physical training induces specific changes in cardiac structure and function, commonly referred to as "athlete's heart" [1], [2]. These changes are generally benign and can include symptoms, as showcased in "Norwegian endurance athlete ECG database" [1], such as increased ventricular wall thickness, dilated chambers, and enhanced cardiac output. However, distinguishing these normal adaptations from pathological alterations that mimic the ECG characteristics of an athletic heart remains a clinical challenge.

The goal of this project is to develop an advanced signal processing methodology that enhances the analysis of ECG data to distinguish normal athletic cardiac remodeling from potential pathological conditions. By employing wavelet transform techniques and heart rate variability (HRV) analysis, the project aims to extract and scrutinize subtle features within the ECG signals that are indicative of specific cardiac conditions or adaptations. The wavelet analysis offers a time-frequency representation of the ECG, providing a powerful framework to capture transient and non-stationary cardiac events that may not be evident in the traditional time-domain analysis.

This study builds upon the foundational understanding that the morphology and timing of ECG waveforms, such as the P-wave, QRS complex, and T-wave, are influenced by physiological factors related to athletic training. By leveraging the multiresolution capabilities of wavelet transforms, the project seeks to preserve the diagnostic integrity of the ECG signal while isolating the noise and artifacts that could obscure critical cardiac features. The processed ECG data was further analyzed to calculate time intervals, such as RR intervals (the time intervals between successive R waves in the QRS complex), which offer valuable information regarding the HRV and autonomic regulation in athletes. HRV is the fluctuation in the time intervals between adjacent heartbeats [3], and will be the main metric to determine whether an athlete has any underlying heart condition or not in this project. This was done by comparing five of these athletes to one another through feature extraction and subsequent analysis of the signal and through HRV calculations.

To further elucidate the components of ECG waveforms, it is necessary to deconstruct the signal into its three essential parts:

- **P-wave**: As illustrated in Figure 1 the P-wave is the short upward movement of the ECG tracing and represents atrial depolarization, which is the elecrical activation that preceds the contraction of the atria [4], [5]. In a healthy individual, the P-wave is smooth and rounded as illustrated in the aforementioned figure. Abnormalities in the P-wave can indicate atrial enlargement, atrial arrhythmias or issues with the electrical conduction system within the atria.
- **QRS complex**: Following the P-wave, the QRS complex is a quick succession of three waves: a downward deflection (Q), a large upward deflection (R), and a downward deflection (S). It represents ventricular depolarization, the electrical activation of the ventricles that causes them to contract and pump out blood to the body and lungs [4], [6], [7]. The QRS complex's shape and duration can be affected by various cardiac conditions, including ventricular hypertrophy, blockages in the electrical conduction

pathways, or myocardial infarction [8].

- **T-wave**: The T-wave is a slightly asymmetric wave that follows the QRS complex and represents ventricular repolarization, the process by which the ventricles recover and prepare for the next cycle of depolarization. The appearance of the T-wave can vary significantly among healty individuals, but abnormalities such as T-wave inversion and peaked T-waves can indicate issues such as elecrolyte imbalances, ischemia, or ventricular hypertrophy [4], [7], [9].

The data used in this study was accessed through an open-access database. The dataset comprised of ECG recordings from 28 elite Norwegian endurance athletes, where 24 participants were rowers (86%), accompanied by two kayakers (7%) and two (7%) cyclists. The gender distribution of the participants was predominantly male with 19 men (68%) and 9 women (32%). The age range of the athletes spanned from 20 to 43 years of age, with an average age of 25 and a standard deviation of 4.7 years. In terms of training volume, the data from 2017 showed an average of 822 training hours, fluctuating by a standard deviation of ±117 hours. The following year saw a similar average of 820 hours, with a slightly reduced variability of ±113 hours. In 2019, the average training duration slightly decreased to 798 hours, with an increased standard deviation of ±171 hours.

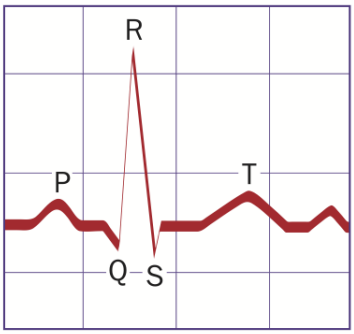


Figure 1. Standard ECG waveform containing P-wave, QRS complex, & T-wave.

Source: https://www.healio.com/cardiology/learn-the-heart/ecg-review/ecg-interpretation-tutorial/p-wave

A. Data Acquisition

For the data acquisition the athletes were lying horizontally on a bench relaxed while 10 electrodes were attached as shown in Figure 2. Each ECG recording encompasses a comprehensive 12-lead capture to measure and record the electrical activity of the heart from twelve different perspectives. These perspectives were captured by placing ten electrodes on the athlete's body; six on the chest (precordial leads) and four on the limbs (limb leads). Each lead represents and acquires data by observing the heart from a different angle, and together provide a comprehensive three-dimensional view of the heart's electrical activity.

The limb leads were placed in a formation known as Einthoven's triangle [10] to give a view of the horizontal plane and is as follows:

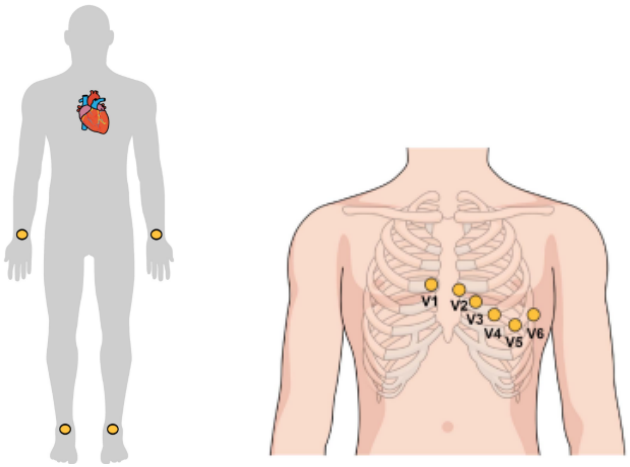
- **Lead I**: Measures the voltage difference between the left arm and the right arm.
- **Lead II**: Measures the voltage difference between the left leg and the right arm.
- **Lead III**: Measures the voltage difference between the left leg and the left arm.

The augmented limbleads also viewed the heart's electrical activity in the horizontal plane but with a central reference point:

- **aVR**: Records the electrical activity from the right arm to the center of the heart.
- **aVL**: Records the electrical activity from the left arm to the center of the heart.
- **aVF**: Records the electrical activity from the left leg or foot to the center of the heart.

V1 – V6: These precordial leads are placed in specific positions on the chest surrounding the heart as illustrated in Figure 2 (b) and provides a view of the heart's electrical activity in the vertical plane. V1 and V2 are placed on either side of the sternum, V4 at the mid-clavicular line in the fifth intercostal space, with V3 between V2 and V4. V5 and V6 follow in line with V4 but under the anterior and mid-axillary lines, respectively.

The recordings were performed as a standard 10 seconds resting ECG and sampled with a sampling frequency of 500 Hz.The device used to record the ECG was a GE MAC VUE 360.



(a) Limb Leads (b) Precordial Leads

Figure 2. Placement of the limb leads & precordial leads on the athletes from the open-access database [1].

II. Related Work

In ECG research the standardized placement of the leads are comparable to the Norwegian study [1] where the data collection was assessed from. The limb leads were placed

on the wrists and ankles as shown in Figure 2 (a), and the precordial leads were attached to the chest as shown in Figure 2 (b), according to Ashley EA et al's book regarding decades of ECG research [4] among other sources [11], [12], [13] & [14].

The study from Li et al. [15] explores the automatic recognition of arrhythmias using a novel multi-domain feature extraction method for ECG signals. The methodology involves an improved wavelet threshold for noise reduction, kernel-independent component analysis for nonlinear feature extraction, and discrete wavelet transform for frequency domain features. This advanced approach to ECG signal processing is then classified using a support vector machine optimized by a genetic algorithm. Their system demonstrated high classification accuracy when applied to the MIT-BIH arrhythmia database, showing its effectiveness in classifying various types of heartbeats which could be crucial for automatic identification of cardiac arrhythmias.

Based on the review of the article "Improved Detection of ECG Features Using Wavelet for Emergency Medical Application" by A.K.M Fazlul Haque [16], the study presents an innovative approach to ECG signal analysis, specifically designed to enhance the detection of subtle ECG features in emergency medical situations. The research leverages wavelet transforms to refine feature extraction from both normal and noise-corrupted ECG signals. The proposed method demonstrates a significant improvement over conventional fast fourier transforms and wavelet techniques, offering a new level of precision in identifying minor abnormalities that are not easily visible in graphical ECG representations.

Aziz, S. et al. [17] presents a methodology that uses wavelet transforms and a fusion algorithm for enhanced ECG signal processing and feature extraction. This method outperforms traditional techniques, especially in emergency medical applications where accurate and prompt diagnosis is critical. Furthermore, it uses a comprehensive database for machine learning, which is unique and realistic for classification tasks, emphasizing the importance of using large, diverse datasets for training models.

Incorporating wavelet transform and machine learning into the acquired ECG database offers an extensive approach in feature extraction and signal analysis to distinguish from a healthy heart or an underlying cardiac disease for the 28 athletes. Wavelet transform, as employed in the studies by Li et al [15] and Haque [16], can be adapted to effectively isolate and enhance the cardiac signatures unique to athletes, particularly for subtle changes induced by intensive traning. Machine learning algorithms, proven in their ability to classify complex patterns within ECG data, can be trained on these refined features to distinguish between benign training effects and potential pathologies.

III. Method

The raw ECG data, initially in .hea and .dat formats for each athlete, were converted to .csv format for simplified data manipulation. This conversion was done by a self-made Python script that employed the WFDB Python library, a library made for handling physiological signals. This step ensured the data was in a readily analyzable format for subsequent processing.
Upon conversion, the ECG data was imported into MATLAB, maintaining the original sampling frequency of 500 Hz to preserve the temporal fidelity of the signals. Preprocessing in MATLAB involved several crucial steps aimed at ensuring the ECG waveforms were accurately represented and primed for feature extraction. Specifically, the preprocessing employed Daubechies wavelet transform, chosen for its compact support and orthogonality, ideal for ECG signal analysis [18]. A decomposition level of 3 was utilized, balancing the need for noise reduction with the preservation of vital signal features.

The R-peak detection process was a pivotal aspect of the analysis, crucial for identifying the ventricular depolarization phase within the ECG data. An automated MATLAB script executed this task across all 12 leads, employing a thresholding mechanism to pinpoint R-peaks based on local maxima that surpassed a set threshold, calculated as the median of the signal plus its standard deviation. This is illustrated in Figure 3. This approach was refined to account for the "athlete's heart" phenomenon by incorporating a refractory period of 350 milliseconds, ensuring T-wave amplitudes did not confound R-peak detection.

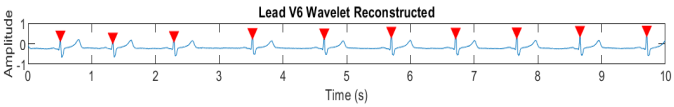


Figure 3. Lead V6 Wavelet Reconstructed for athlete 1.

Subsequent to R-peak identification, RR intervals were computed, laying the groundwork for HRV analysis. The HRV metrics were then derived directly from these intervals, encompassing mean RR intervals, SDNN, RMSSD, NN skewness, kurtosis, among others. These parameters were chosen for their capacity to collectively provide a comprehensive overview of the autonomic nervous system's influence on heart function, particularly in the context of athletic training and recovery. These are discussed later in this section and in Section IV & Section V.

The HRV parameters were calculated in MATLAB from the raw ECGs using a code implementation that the authors made.

Measures of normal RR-intervals (NNs) rate used:

- **MeanRR** (ms) - The average of all NN intervals (the time between successive normal heartbeats) measured in milliseconds. A lower MeanRR could indicate a higher average heart rate, and vice versa.

- **NNskew** - Skewness is a statistical measure that describes the asymmetry of the distribution of NN intervals. Skewness can provide insights into how the NN intervals deviate from a normal distribution, potentially indicating arrhythmias or other irregularities in heart rhythm.
- **NNkurt** - Kurtosis measures the leptokurticity or "peakedness" of the distribution of NN intervals. High kurtosis values indicate a distribution with heavy tails and a sharper peak, suggesting more frequent extreme deviations in heart rate variability.

Measures of NN intervals variability used:

- **NNiqr** (ms) - The Interquartile Range (IQR) measures the spread of the middle 50% of NN intervals. It is calculated as the difference between the 75th and 25th percentiles of the NN interval distribution. A wider IQR suggests greater variability in the heart rate over the measurement period.
- **SDNN** (ms) - The Standard Deviation of NN intervals (SDNN) quantifies the overall variability in NN intervals, reflecting all the cyclic components influencing heart rate variability within the period of analysis. Higher SDNN values indicate greater overall heart rate variability.
- **RMSSD** (ms) - The Root Mean Square of the Successive Differences between NN intervals (RMSSD) specifically measures the short-term variability in heart rate. It is particularly sensitive to high-frequency changes in heart rate, which are predominantly influenced by the parasympathetic nervous system. Higher RMSSD values are often interpreted as indicative of increased vagal tone.
- **pNN50** (%) - This metric represents the percentage of pairs of successive NN intervals that differ by more than 50 milliseconds. It is another measure of short-term variability and is thought to reflect parasympathetic activity. A higher pNN50 value suggests more frequent significant changes in heart rate.
- **LF**(ms^2) - Low Frequency: Reflects the power within the low-frequency band (0.04 to 0.15 Hz) of the HRV spectrum. LF represents a mixture of sympathetic and parasympathetic activity, with an emphasis on sympathetic modulation. Higher LF values indicate increased sympathetic activity or reduced vagal activity, especially when considered relative to HF.
- **HF**(ms^2) - High Frequency: Measures the power within the high-frequency band (0.15 to 0.4 Hz) and is primarily associated with respiratory sinus arrhythmia, thus reflecting parasympathetic (vagal) influences on the heart rate. Higher values of HF suggest greater vagal modulation.
- **LF_Norm** - This is the normalized value of LF power, calculated as a percentage of the total power minus the very low-frequency component. LF_Norm is used to assess the balance of sympathetic to vagal tone, with higher values suggesting a predominance of sympathetic modulation over parasympathetic influences.
- **HF_Norm** - Represents the normalized value of HF power, also calculated as a percentage of the total power minus the very low-frequency component. HF_Norm indicates the balance of vagal tone relative to the overall autonomic modulation. Higher values suggest a dominance of parasympathetic activity and can be associated with relaxation and decreased stress.

For this project, five athletes (numbers 1, 9, 15, 21, and 28) were chosen for analysis. The selection was described as "random" and "hand-picked," which suggests a non-systematic approach to sampling from the larger group of 28 athletes. This method of selection could introduce bias, and the findings from this subset may not be generalizable to all athletes in the study or the broader athlete population.

Lead II was chosen for feature extraction and HRV analysis, aligning with common practices in HRV research. Lead II, formed by the electrodes placed on the right arm and left leg, provides a clear view of the heart's electrical activity, particularly the QRS complex, making it suitable for detecting R-peaks and calculating NN intervals. This choice is supported by literature, as referenced [19], [20], & [21], highlighting its widespread use and acceptance in HRV studies.

IV. Results

By performing the HRV analysis and running the MATLAB code on athlete 1, 9, 15, 21, & 28 respectively a capture of the results are in Table I which are the key findings from the HRV analysis.

Table I
METRICS FOR THE RESPECTIVE ATHLETES

	Athlete 1	Athlete 9	Athlete 15	Athlete 21	Athlete 28
MeanRR (s)	1.0220	1.0440	1.0762	1.1931	1.0871
SDNN (ms)	0.1150	0.1193	0.1251	0.0920	0.1007
RMSSD (ms)	0.1218	0.1307	0.1073	0.0784	0.1203
NNskew	0.1640	-2.2408	-2.0287	-0.3247	-1.3878
NNkurt	2.9756	6.4501	5.8405	1.8574	3.1442
NNiqr (ms)	0.1140	0.0580	0.0455	0.1340	0.0590
pNN50 (%)	75	25	62.5	83.3333	25
LF (ms^2)	0.0056	0.0038	0.0074	0.0060	0.0025
HF (ms^2)	0.0099	0.0087	0.0085	0.0049	0.0055
LF_Norm	0.3612	0.3047	0.4653	0.5480	0.3154
HF_Norm	0.6388	0.6953	0.5347	0.4520	0.6846

Athlete 21 displayed the longest mean RR interval duration (1.1931 s), indicating a slower average heart rate, while Athlete 1 had the shortest mean NN interval duration (1.0220 s), suggesting a relatively faster heart rate.

In terms of overall variability, Athlete 15 exhibited the highest SDNN value (0.1251 ms), reflecting greater variability in heart rate compared to the other athletes. Athlete 9 demonstrated the highest RMSSD value (0.1307 ms), indicating greater parasympathetic activity. The skewness of the NN intervals distribution was notably negative for Athlete 9 (-2.2408), suggesting a left-tailed distribution. Additionally, Athlete 9 exhibited the highest kurtosis value (6.4501), indicating a peaked distribution with heavy tails.

Regarding the spread of the middle 50% of the data, Athlete 21 showed the highest interquartile range of NN intervals (NNiqr) value (0.1340 ms), reflecting a wider range of NN intervals compared to the other athletes. Athlete 21 also demonstrated the highest percentage of adjacent NN intervals differing by more than 50 ms (pNN50) value (83.33%), indicating greater parasympathetic activity. In terms of sympatho-vagal balance, Athlete 3 exhibited the highest power in the low-frequency (LF) band (0.0074 ms^2), while Athlete 2 exhibited the highest power in the high-frequency (HF) band (0.0087 ms^2). Additionally, Athlete 21 exhibited the highest normalized LF power (LF_Norm) value (0.5480), indicating a predominance of sympathetic activity.

V. Discussion & Conclusions

The HRV analysis of the five elite athletes provided insightful observations into their autonomic function and cardiovascular efficiency. For instance, Athlete 1's shorter mean RR interval suggests a relatively higher heart rate at rest, potentially indicating a lower level of vagal tone or cardiovascular efficiency. Such findings could be juxtaposed with existing literature that often associates high-intensity training with significant autonomic adaptations, leading to increased vagal modulation and HRV in athletes [3]. With a relatively lower LF (0.0056 ms²) and higher HF (0.0099 ms²) alongside a balanced LF_Norm (0.3612) and HF_Norm (0.6388), may indicate a well-maintained autonomic function, with a slight inclination towards parasympathetic activity. This could reflect good recovery status and cardiovascular efficiency, contrasting with their shorter mean RR interval which suggests a higher resting heart rate.

Athlete 9's HRV profile, characterized by the highest RMSSD, points towards a pronounced parasympathetic activity, aligning with research suggesting that enhanced parasympathetic modulation is a hallmark of athletic training adaptation (Plews et al., 2013). However, the negative skewness and high kurtosis observed in Athlete 9's HRV distribution warrant further investigation, as they suggest a more complex autonomic response that could have implications for training and recovery strategies [22]. This athlete also shows a distinctive autonomic profile with the lowest LF (0.0038 ms²) and a high HF (0.0087 ms²), combined with lower LF_Norm (0.3047) and higher HF_Norm (0.6953), indicating a strong parasympathetic dominance. This profile, consistent with high RMSSD values, underscores pronounced vagal modulation, which aligns with adaptive responses to endurance training.

Athlete 15's HRV analysis indicates a substantial SDNN which suggests a well-regulated autonomic nervous system, hinting at a broad HRV range associated with a healthy cardiovascular system. The output for athlete 15 the elevated T-wave amplitude relative to the R-peak was observed, indicative of this patient having symptoms of athlete's heart [2]. Furthermore, this athlete exhibits the highest LF (0.0074 ms²), suggesting enhanced sympathetic activity. This is coupled with a moderate HF (0.0085 ms²). Their LF_Norm (0.4653) is also elevated relative to others, implying a balance tilted towards sympathetic drive. Yet, their substantial SDNN indicates a broad range of HRV, typically a sign of a healthy cardiovascular response.

Athlete 21's HRV profile, distinguished by the longest mean RR interval, further exemplifies these adaptations, showcasing high cardiovascular fitness typical of endurance athletes. The longer RR intervals indicate a slower heart rate, reflective of an efficient heart function. However, Athlete 21's lower RMSSD and SDNN may suggest reduced beat-to-beat variability, highlighting a potential area for enhancing autonomic balance and refining recovery strategies. This athlete showcased the second-highest LF (0.0060 ms²) and lower HF (0.0049 ms²) values, along with the highest LF_Norm (0.5480), which reflects a notable sympathetic predominance.

Athlete 28's metrics suggest a balance between sympathetic and parasympathetic activity, with HRV values that do not show extreme deviations from the group's norms. This could indicate a well-adapted cardiovascular system, but the lowest values for both LF (0.0025 ms²) and HF (0.0055 ms²), as well as for LF_Norm (0.3154) and HF_Norm (0.6846) could suggest a more balanced autonomic state but with a potential lean towards parasympathetic dominance.

The Norwegian athlete database [1] did not specify the sports in which the athletes participated, nor did it provide personal details such as gender or age. Consequently, it was not possible to evaluate the specific impacts of sports such as rowing, kayaking, and cycling. Additionally, it is important to reiterate that the data collection was conducted while all the athletes were at rest.

The development and refinement of the MATLAB script for data preprocessing, filtering, and feature extraction for HRV analysis were intricate and time-intensive endeavors. Leveraging existing libraries, such as those provided by the PhysioNet Cardiovascular Signal Toolbox [23], [24] could have streamlined this process, reducing the time spent on scripting and potentially offering a more precise depiction of the data. Having the scripts being self-made could have yielded more possible room for error. A study with similar

aim and methodology utilized this toolbox to streamline their process [25] which proved to be a sufficient methodology.

One significant constraint of this study was its narrow concentration on a limited group of five elite athletes, that could argue to have been picked with bias as mentioned in Section III. Broadening the research to encompass a comparative evaluation alongside individuals possessing standard cardiovascular profiles could have deepened the insights, accentuating the unique physiological characteristics of elite athletes in contrast to the general population.

VI. Acknowledgement

The authors would like to thank examiner Ivan Tomasic for their help and support during the project; providing insightful information and tools to be able to tackle this project head on.

VII. Author Contributions

Both authors, Mattias Eek & Marcus Andersson, equally and collaboratively contributed to all facets of this project. Being only two contributors made it simple to make a joint effort and work together in all aspects of the project.

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